G Sudarshan Sastry

Full-Stack AI Engineer | LLM & RAG Systems

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Professional Summary

Full-Stack AI Engineer shipping LLM, RAG and voice-agent systems end to end: FastAPI/Django backends, React/Next.js frontends, PostgreSQL/Redis and Dockerised deployments on AWS, GCP and OCI. Delivered 14+ LLM/RAG systems for 9 clients, from schema design to production. First-author IEEE IGARSS 2025 researcher with an embedded-systems and robotics background.

Core Skills

Languages: Python, JavaScript/TypeScript, C, Java, Rust, SQL
Backend & Web: FastAPI, Flask, Django, REST API Design, Next.js, React, Node.js, Prisma, WebSockets, WebRTC, Pipecat
AI / LLM: RAG, FAISS, Qdrant, pgvector, LangChain, LLM Routing, LLM-as-Judge Evals, Gemini/OpenAI/Claude APIs, Ollama, PyTorch, TensorFlow, scikit-learn
Infra & Data: Docker, Linux, AWS/GCP/OCI, Cloud Run, Vercel, PostgreSQL, Supabase, Redis, MongoDB, SQLite, Git
Automation & Robotics: n8n, Pabbly, Blue Prism RPA, Webhooks; ROS, Gazebo, Unity3D, KiCad, STM32, MATLAB

Experience

- **Associate AI Developer** (AI Intern, Nov 2025; converted Jan 2026; full-time Apr 2026) *Remote*
  Freedom With AI Nov 2025 – Jul 2026
  - **Delivered 14+ LLM/RAG systems across 9 client accounts** with provider-agnostic LLM routing (retry/fallback), vector retrieval, FastAPI + Next.js services and n8n event orchestration, owning requirement to deployment.
  - **Productised a webinar & call attribution platform** (Next.js, PostgreSQL/JSONB) reconciling 4 mismatched export sources into one schema across 7 migrations; one codebase redeployed to 5 tenants.
  - **Ran a causal study on an AI calling channel across 100+ webinars and 5 cohorts**: purchase lift of +20–70% vs +4–5% on attendance; conversations beyond 3 turns predicted a 7.8% purchase rate while call connection predicted nothing.
  - **Recovered ∼37,000 validated CRM contacts and 61 automations** by reverse-engineering an undocumented internal API (XHR endpoint mapping, auth-flow replay, bypassing a silent 5,000-row truncation).
  - **Shipped an AI tool suite on Cloud Run** (Next.js, 124 commits): prompt enhancer, AI tool finder, job-prep assistant, Chrome extension and a Claude-based assignment auto-grader.
  - **Operated 16+ Dockerised services on a six-VM AWS/GCP/OCI fleet** (43 Dockerfile/Compose artifacts) with SSH runbooks; distilled a 49,737-message mail corpus into a ∼5,000-pair support knowledge base; ran KT for ∼12 colleagues.
- **Software Engineer** (Intern from Mar 2025) *Hyderabad, India*
  Deep Algorithms Mar 2025 – Sep 2025
  - **Built a DDoS detection path** with packet-to-flow aggregation computing per-source packet rate, source-IP entropy and protocol-mix features, evaluated against tunable thresholds for volumetric and protocol-layer attacks.
  - **Implemented TLS fingerprinting from ClientHello metadata** (cipher suites, extensions, version) to key detections on client-stack identity rather than source IP; owned feature specs and acceptance tests with QA.
- **Research Intern** *Mandi, India*
  CAIR Lab, IIT Mandi Dec 2024 – Mar 2025
  - **Wrote STM32 BLDC motor firmware in C** with timer-driven PWM commutation and closed-loop speed control, from bare-metal bring-up to bench validation.
  - **Fused multi-sensor gas readings on Arduino** using per-sensor calibration curves, drift compensation and noise filtering; authored firmware docs and test plans adopted by the lab.
- **AI-ML Intern** *Hyderabad, India*
  AKOSCOWL DOT INVENTIONS Pvt. Ltd. Feb 2024 – Sep 2024
  - **Designed drone power and control electronics in KiCad** and built a ROS/Gazebo model (URDF airframe, sensor plugins, flight-controller-in-the-loop).
  - **Developed obstacle-avoidance planners** emitting MAVLink waypoints through QGroundControl mission flows, validated in simulation before the flight-test campaign.

Other Internships

- **Python Development Intern — CodeClause** *Jul 2023 – Aug 2023*
  Voice assistant, plagiarism checker, URL shortener, music player; awarded a Letter of Recommendation.
- **Machine Learning Intern — Bharat Intern** *Dec 2023 – Jan 2024*
  House-price prediction (XGBoost), TF-IDF movie recommender, Iris classification.
- **Data Engineering Virtual Internship — AICTE-EduSkills, AWS Academy** *Sep 2023 – Nov 2023*
- **Intelligent Automation Virtual Internship — AICTE-EduSkills, SS&C Blue Prism** *May 2023 – Jul 2023*
- **AI with Python Intern — Coincent** *May 2023 – Jun 2023*
  Landmark-detection project, after a 2-month AI with Python training program.
- **Python Development Virtual Intern — Coders Cave** *1 month*
  Music player, calculator and two other GUI apps.
- **AI-ML Virtual Internship — AICTE-EduSkills, AWS Academy** *Jul 2022 – Sep 2022*

Products and Projects

- **NeuroWrite — Human-Like Handwriting Generation (Independent Research)** *2026*
  PyTorch | Diffusion Models | Distributed Training | Apple MLX
  - **Modelled structured intra-writer variation** with a diffusion model: writer style as a distribution $N(\mu, \Sigma)$, a neuromotor-inspired noise schedule and a repetition-memory cross-attention module.
  - **Proposed the Writer Variation Fidelity metric** (1 − Wasserstein-1 over pairwise distance distributions); trained on a self-assembled pool of 3 Kaggle + 3 Lightning AI free-tier accounts.

- **UAV Obstacle Avoidance with Deep RL — IEEE IGARSS 2025 (First Author)** *2025*
  Python | Unity3D | DQL | DDPG
  - **Trained DQL and DDPG agents in Unity3D** comparing 2D LiDAR vs depth estimation for fixed-wing UAVs; LiDAR won for real-time control, hybrid sensing proposed for production.

- **Match Check — AI Deal-Breaker Detector for Matchmakers** *Oct 2026*
  JavaScript | Claude API | Node Test Runner
  - **LLM reads, rules decide**: Claude extracts deal-breakers from free-text bios (exact quote or null), then a deterministic engine scores (multiplicative deal-breakers, weighted fit, geometric mean of both sides) with every verdict traceable.

- **AI Intelligent Assistant Platform — Voice Agent**
  Python | FastAPI | Docker | PostgreSQL | Redis | WebRTC | Pipecat | Gemini API
  - **Built a multi-service AI assistant** (APIs, NLP, real-time WebRTC voice) whose agent places/receives calls, sends WhatsApp and email, handles approvals, keeps user memory and generates daily briefings.

- **Content Engine — Multi-Tenant Script Generation**
  FastAPI | React | Qdrant | SearXNG | n8n | GCP | Oracle Cloud
  - **Combined vector retrieval over a research corpus with live search and trend ingestion** for templated script generation; evolved from n8n workflows into a standalone service deployed on two clouds.

- **AppFlowy-AI — Local-First AI Workspace**
  Flutter | Rust | SQLite | Ollama | Vector Search
  - **Built an offline AI workspace**: SQLite canonical store, two-way YAML projections, filesystem watchers for conflict-free edits, local vector search and an agent router dispatching to Ollama models.

- **AI Story Writing Engine — RAG Platform**
  React | Flask | MongoDB | FAISS | Azure OpenAI | WebSockets | Vosk STT
  - **Built a full-stack AI writing platform** with RAG (FAISS + Azure OpenAI), intent-based story/chat routing, WebSocket speech-to-text and persistent conversation history.

- **Instagram Content Automation Pipeline**
  n8n | Gemini / OpenAI APIs | Instagram Graph API | Cron
  - **Automated news-to-post publishing**: RSS ingestion, AI idea/caption/image generation, approval gate, media queue and retry/backoff; 100+ posts/month published unattended.

- **Multimodal Emotion-Aware Conversational AI**
  Flask | AssemblyAI | Wav2Vec2 | Keras | librosa | OpenAI
  - **Combined STT, speech-emotion recognition** (constant-Q harmonic coefficients + Keras classifier, Wav2Vec2) and LLM responses with TTS fallback and session memory.

Education

- **B.Tech, Electronics and Communication Engineering — CGPA 8.59/10** *Hyderabad, India*
  KL University — Specialization in Data Science & Big Data Analytics Aug 2021 – Aug 2025

Certifications

- **Microsoft:** Azure AI Fundamentals (AI-900) **Oracle:** OCI 2023 Data Science Professional; OCI 2023 Foundations Associate **Salesforce:** AI Associate
- **SS&C Blue Prism:** Associate Developer **MongoDB:** Python Developer Path **Aviatrix:** Multicloud Network Associate **Pathfinder:** Advanced & Essentials Automation

Achievements

- **First-Author Publication — IEEE IGARSS 2025:** Obstacle Avoidance in Drones using Deep Reinforcement Learning (DOI 10.1109/IGARSS55030.2025.11243721).
- **1st Place — JHUB-JNTUH Regional Hackathon** (Jun 2022) · **Languages:** English, Telugu, Hindi
- **Selected for Amazon ML Summer School 2023** · **Official Coding Mentor**, Department of ECE, KL University.
- **Four Letters of Recommendation** — CodeClause and three KL University faculty, including the ECE Head of Department.